

RELTWIN: CONTRASTING LOCALLY VALID TIMESTAMP ANSWERS FOR QUERY-SPECIFIC AUDIO GROUNDING

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ABSTRACT

Audio temporal grounding maps a text query to the corresponding event intervals. When the same sounds recur in different orders, event recognition alone cannot determine which occurrence satisfies a relational query: multiple windows are acoustically valid, yet only one matches the requested relation. We introduce RelTwin, a fine-tuning framework that contrasts such locally valid timestamp answers. Shared sound excerpts form opposite-order windows, and inverse queries exchange their correct targets. RelTwin first learns query-dependent occurrence selection through complete-answer competition, then ranks interval sets by coverage, count, and boundary accuracy. Across SpotSound-Bench, Clotho-Moment, and UnAV-100, it improves SpotSound-A and TimeAudio by up to 4.23 and 14.48 mIoU points, respectively. On 300 inverse-query pairs, SpotSound-A candidate supervision raises joint accuracy from 55.33 to 81.67; TimeAudio also improves over paired SFT. Separate silence interventions support SpotSound-A’s selection gains. Inference retains direct timestamp generation.

Index Terms— Audio temporal grounding, timestamp answer contrasts, query–window binding, paired supervision

1. INTRODUCTION

Audio temporal grounding maps a natural-language query to event intervals in a recording [1, 2]. When sound categories repeat, recognition alone cannot determine which occurrence satisfies the query. A recording may contain both “dog followed by rooster” and “rooster followed by dog,” which inverse queries should map to different windows. Returning the same window for both is a *query–window binding error*: the prediction is acoustically supported but relationally incorrect.

Prior work advances several parts of audio grounding. Weakly supervised methods learn localization from clip-level captions [3], while audio-language models support open-ended generation [4, 5, 6]. TimeAudio adds temporal markers

and absolute time-aware encoding [7]; SpotSound interleaves timestamps with audio tokens and trains on absent-event queries [8]. These systems improve timestamp generation, but their positive sequence objectives do not explicitly compare two acoustically valid answers.

Compositional audio–language methods distinguish alternative descriptions through audio–text alignment [9], counterfactual captions [10], temporal contrast [11], composition-aware negatives [12], and multi-grain alignment [13]. SHINE constructs negative queries and saliency rankings for video grounding [14]. RelTwin instead scores complete timestamp answers for coexisting windows whose correctness exchanges with the query.

The challenge is to distinguish two *locally valid timestamp answers*, not to construct an obviously incorrect negative. Positive sequence supervision raises the correct answer’s likelihood without requiring it to outrank another acoustically supported window. A controlled contrast should hold the audio fixed, change the queried relation, and compare complete interval sets because errors can involve occurrence, coverage, count, or boundaries.

RelTwin (Fig. 1) introduces this competition through two-stage fine-tuning. Shared excerpts form opposite-order windows whose targets exchange under inverse queries. Stage 1 contrasts complete serialized answers and couples their preferences under reversal; Stage 2 compares interval sets with different coverage, counts, and boundaries. Timestamp-answer scores and localization replay adapt SpotSound-A and TimeAudio while preserving direct generation without candidate scoring at inference.

Both backbones improve on three public benchmarks and the 300-pair inverse-query evaluation. SpotSound-A candidate supervision reduces identical answers from 61 to 2; two-stage RelTwin reduces TimeAudio’s from 160 to 34. A separate SpotSound-A intervention retains the candidate-supervision advantage after changing silence timing while preserving event waveforms, order, and queries.

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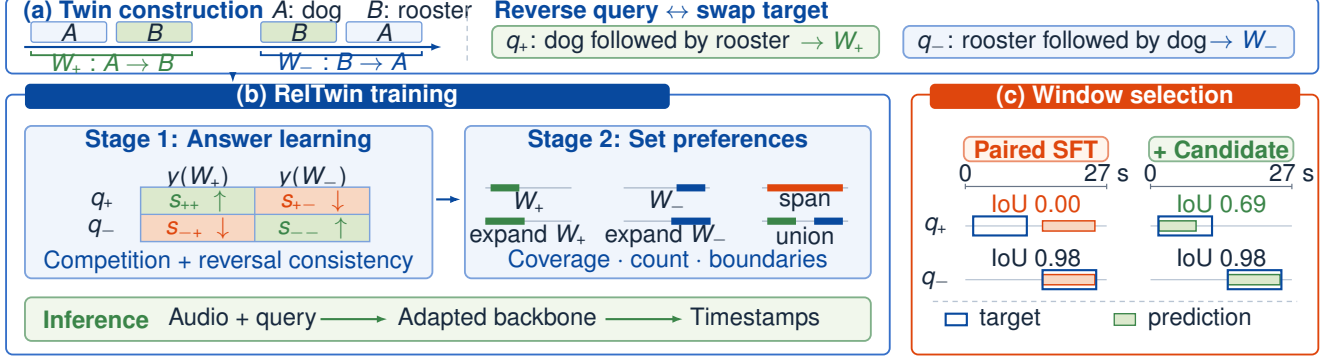


Fig. 1. RelTwin. (a) Shared excerpts form opposite-order windows $W_{\pm}$ for queries $q_{\pm}$. (b) Green/orange cells mark matching/competing timestamp answers; arrows indicate the desired score changes. Stage 2 compares interval sets. (c) SpotSound-A paired SFT versus candidate supervision: outlines are targets, filled bars are predictions, and numbers are set-IoU.

2. METHOD

RelTwin organizes supervision around competition between locally valid timestamp answers (Fig. 1). Section 2.1 constructs query-dependent twins from shared excerpts and inverse queries. Section 2.2 uses complete-answer competition (Stage 1) to select the queried window and reverse the preference with the query. Section 2.3 introduces quality-aware interval-set preferences (Stage 2) for coverage, count, and boundary errors. For recording x of duration T and query q , the output is $S = \{[a_k, b_k]\}_{k=1}^n$, where $0 \leq a_k < b_k \leq T$.

2.1. Query-dependent twin construction

This construction supplies the competing examples used by both training stages. To make the query determine window selection, we arrange the same ESC-50 excerpts [15], A and B , as $A \rightarrow B$ and $B \rightarrow A$ in one recording. Targets $W_{\pm}$ span each sequence, including its gap; queries $q_{\pm}$ request the respective orders. Both windows contain the same categories, but W_{-} is negative for q_{+} and positive for q_{-} . We alternate which order appears first.

2.2. Stage 1: complete-answer competition

Stage 1 establishes query-specific selection by comparing complete answers under each query. We serialize S as comma-separated spans for SpotSound-A or temporal tokens for TimeAudio. Answer tokens $y(S)$ exclude the prompt and receive the mean log-probability

$$s_{\theta}(x, q, S) = \frac{1}{|y(S)|} \sum_t \log P_{\theta}(y_t | x, q, y_{<t}). \quad (1)$$

The 2×2 matrix $M_{ij} = s_{\theta}(x, q_i, W_j)$ indexes queries by rows and answers by columns, $i, j \in \{+, -\}$ (Fig. 1b). Its diagonal entries are the correct query-answer pairs. Sequence supervision minimizes $\mathcal{L}_{\text{seq}} = -(M_{++} + M_{--})/2$, rewarding each correct answer independently. Candidate supervision adds the missing competition by training each diagonal

score to exceed the other acoustically valid answer in its row. For $p_i(j) = \text{softmax}_j(M_{ij}/\tau)$ with $\tau > 0$,

$$\mathcal{L}_{\text{cand}} = -\frac{1}{2}[\log p_{+}(+) + \log p_{-}(-)]. \quad (2)$$

Query reversal should also reverse the answer preference. We align the two distributions after exchanging candidate labels using the Jensen–Shannon divergence $\mathcal{L}_{\text{JS}} = \text{JS}(p_{+}, \Pi p_{-})$, where Π swaps the candidates. Replay retains ordinary localization supervision through examples (x_r, q_r, S_r) with $\mathcal{R}_1 = -s_{\theta}(x_r, q_r, S_r)$. Stage 1 optimizes

$$\mathcal{L}_1 = \mathcal{L}_{\text{seq}} + \lambda(\mathcal{L}_{\text{cand}} + \gamma \mathcal{L}_{\text{JS}}) + \beta \mathcal{R}_1. \quad (3)$$

2.3. Stage 2: quality-aware interval-set preferences

Stage 2 assigns graded preferences to imperfect answers according to coverage, interval count, and boundaries, rather than treating them as equally wrong. Its six candidates are $W_{\pm}$, their expanded versions, their enclosing span, and their union. Expansions expose boundary excess; the span includes intervening background; the union adds the competing occurrence.

For candidate S_j and target W_i , a four-axis quality vector h_{ij} measures set-IoU, threshold margin, count agreement, and boundary accuracy. The threshold axis averages soft F1 at $\eta = 0.3/0.5$. Each interval’s best-match IoU u contributes $\max(0, u - \eta)/(1 - \eta)$; averaging over predictions and targets gives soft precision and recall. With n_S denoting interval count, count agreement is $1 - |n_{S_j} - n_{W_i}| / \max(n_{S_j}, n_{W_i}, 1)$. Boundary accuracy is $\exp(-e)$: e averages the summed endpoint error for each target’s best-IoU match, divided by $2 \max(b - a, 0.1)$ for target $[a, b]$.

Among K candidates, let d_{ij}^{+} and d_{ij}^{-} count those dominated by and dominating S_j . Dominance means no worse on any axis and better on at least one. We define utility as

$$v_{ij} = \frac{1}{2} \left[\frac{1}{4} \sum_{\ell=1}^4 h_{ij\ell} + \frac{d_{ij}^{+} - d_{ij}^{-}}{K - 1} \right]. \quad (4)$$

Mean quality preserves graded differences; dominance rewards improvements without sacrificing another axis. Soft targets $t_i = \text{softmax}(v_i/\tau_q)$ assign higher probability to better interval sets. The same answer scores, normalized over all six candidates, learn this distribution through $\mathcal{L}_{\text{set}} = -\frac{1}{2} \sum_{i,j} t_i(j) \log p_i(j)$. The six-way permutation $\Pi = (2, 1, 4, 3, 5, 6)$ swaps targets and expansions, fixing the span and union.

Replay candidates perturb boundaries, add/drop events, merge spans, split the longest interval, or cover the recording. The same quality rule defines their listwise loss $\mathcal{L}_{\text{set},r}$. On replay examples, the frozen base model supplies a reference distribution p_0 ; the KL constraint anchors preferences p_r to this reference to help retain existing localization preferences. Stage 2 optimizes

$$\begin{aligned} \mathcal{L}_2 &= \mathcal{L}_{\text{seq}} + \lambda(\mathcal{L}_{\text{set}} + \gamma\mathcal{L}_{\text{JS}}) + \beta\mathcal{R}_2, \\ \mathcal{R}_2 &= \mathcal{R}_1 + \lambda\mathcal{L}_{\text{set},r} + \kappa \text{KL}(p_0||p_r). \end{aligned} \quad (5)$$

Both backbones share Eq. 1. Training updates each backbone’s LoRA adapters and bridge modules. Inference uses both adapted components for direct timestamp generation from (x, q) , without twin construction or candidate scoring.

3. EXPERIMENTS

3.1. Experimental setup

Datasets. We evaluate general localization on SpotSound-Bench (400 queries, 387 audios) [8], Clotho-Moment (6,649 queries), and the UnAV-100 subset (100 queries, 77 audios) [2]. For controlled relation learning, 40 ESC-50 classes produce 256 training recordings, 1,024 relation queries (512 pairs), and 512 category queries. The templates “followed by” and “and then” give two inverse-query pairs per recording. Table 2 evaluates 300 inverse-query pairs (600 queries). The separate timing study uses the original development set: 10 held-out classes and 80 recordings, with classes and original Freesound IDs disjoint from adaptation data.

Evaluation. Public benchmarks use set-mIoU and R1 at IoU 0.3/0.5, with predictions clipped to the recording duration. For inverse-query pairs, let $u_{ij} = \text{IoU}_{\text{set}}(\widehat{W}_i, W_j)$, where $\text{IoU}_{\text{set}}(S, G) = |S \cap G|/|S \cup G|$ and an empty union scores zero. PairAcc requires $\min(u_{++}, u_{--}) \geq 0.5$; a swap occurs if $u_{++} \leq u_{+-}$ or $u_{--} \leq u_{-+}$, including ties. Joint-PairAcc requires PairAcc without a swap. Inspired by paired compositional evaluation [16], it separates selection from coverage: the union of two disjoint targets can pass PairAcc but fails JointPairAcc. These are pair-level percentages; same-answer counts require identical interval lists.

Training and implementation. Both backbones train for 384 Stage 1 and 128 Stage 2 updates, with LoRA [17]/bridge learning rates of $2 \times 10^{-6}/5 \times 10^{-6}$. Trainable parameters total 31.76M for TimeAudio and 37.63M for SpotSound-A; inference loads adapters and bridges. Candidate and ex-

change/JS weights are $\lambda = \gamma = 1$, replay weight $\beta = 0.75$, and reference-KL weight $\kappa = 0.2$; $\tau = 1$, $\tau_q = 0.15$. Interval expansion is 15% (minimum 0.05 s). Decoding is greedy, with one beam and at most 128 new tokens.

In the TimeAudio training protocol, category queries target both occurrences of A or B ; category:relation pairs enter the paired objective at 3:1. Replay mixes Clotho-Moment training [2], synthetic scale/cardinality data, TEMPO training [18], and AudioGrounding training [1] at 4:3:2:1.

The SpotSound-A comparison in Table 2 isolates Stage 1 competition. Paired SFT uses $\lambda = 0$ and candidate supervision $\lambda = 1$, both with $\gamma = 0$ and $\beta = 0.5$. They share initialization, data, order, and 256 AdamW [19] updates at 5×10^{-6} , with zero weight decay and gradient clipping 1. LoRA has rank 8, $\alpha = 16$, dropout 0.1, and 20.19M trainable parameters. Each update accumulates one inverse-query pair and one replay query. Candidate supervision adds four gradient-free answer scores and weighted gradient recomputation to SFT’s two positive-answer backpropagations. All variants use 16 kHz mono audio, bfloat16, and direct timestamp generation.

For the timing intervention, we recompose 80 development recordings with a 1.5 s lead-in, 0.25 s within-window gaps, 3 s between windows, and a 2 s tail while preserving waveforms, order, and queries. Frozen matched models evaluate one template (160 queries). Timing changes for 40 BA-first recordings; 40 AB-first recordings remain identical.

Both backbones’ main results and matched point estimates use seed 0. Table 2 reports single-checkpoint results; the TimeAudio comparison uses 384+128 updates for paired SFT and RelTwin. Timing-study CIs use 20,000 source-group bootstrap resamples from the original development grouping (39 connected groups; largest: 18 recordings).

3.2. Grounding performance across benchmarks

RelTwin improves both backbones on all three benchmarks (Table 1). TimeAudio gains 8.16, 5.48, and 14.48 mIoU points; SpotSound-A gains 1.53, 1.08, and 4.23. R1 rises at both thresholds on every dataset, and adapted SpotSound-A leads every listed column. Gains with span-style and temporal-token outputs support complete-answer supervision across timestamp representations.

3.3. Inverse-query selection across backbones

On 300 inverse-query pairs (Table 2), SpotSound-A’s 256-update comparison isolates candidate supervision. Relation mIoU rises from 76.08 to 87.70 and JointPairAcc from 55.33 to 81.67; swap error falls from 36.33 to 7.00 and identical-answer pairs from 61 to 2.

TimeAudio compares paired SFT with complete two-stage RelTwin after 384+128 updates. Relation mIoU rises from 38.89 to 43.86 and JointPairAcc from 2.67 to 10.67;

Table 1. Temporal grounding across three benchmarks with RelTwin adaptation of TimeAudio and SpotSound-A (%; higher is better). Baseline results: SpotSound [8, Table 3]. Δ : gain over the corresponding backbone. Bold marks column maxima.

Method	SpotSound-Bench			Clotho-Moment			UnAV-100 subset		
	R1@.3	R1@.5	mIoU	R1@.3	R1@.5	mIoU	R1@.3	R1@.5	mIoU
<i>Non-LLM models</i>									
WTATG	53.0	34.5	38.4	12.1	6.3	9.1	53.0	37.0	38.4
AM-DETR	25.5	15.0	19.5	89.8	88.0	80.9	59.0	46.0	42.8
<i>Proprietary models</i>									
Gemini-2.5-Flash	34.8	27.8	25.7	45.7	33.7	36.9	48.0	37.0	35.6
Gemini-2.5-Pro	20.0	17.3	19.7	40.4	32.5	32.5	39.0	35.0	34.6
<i>Open-source models and RelTwin adaptation</i>									
Kimi-Audio	3.0	1.3	3.2	0.7	0.1	0.9	8.0	2.0	5.3
Qwen2-Audio	5.5	1.8	4.4	6.1	1.3	5.7	14.0	4.0	9.7
Audio Flamingo 3	8.8	2.8	7.6	32.9	21.8	22.6	35.0	24.0	25.0
TimeAudio	7.8	1.8	10.2	39.5	24.9	28.6	21.0	7.0	16.0
TimeAudio + RelTwin	24.25	13.25	18.36	45.63	30.92	34.08	44.00	25.00	30.48
Δ vs. TimeAudio	+16.45	+11.45	+8.16	+6.13	+6.02	+5.48	+23.00	+18.00	+14.48
SpotSound-A	76.5	59.5	57.9	93.4	91.0	85.6	86.0	74.0	69.8
SpotSound-A + RelTwin	78.00	62.00	59.43	93.61	91.01	86.68	89.00	79.00	74.03
Δ vs. SpotSound-A	+1.50	+2.50	+1.53	+0.21	+0.01	+1.08	+3.00	+5.00	+4.23

Table 2. Inverse-query evaluation on 300 pairs (600 queries). SpotSound-A compares paired SFT and candidate supervision after 256 updates; TimeAudio compares paired SFT and RelTwin after 384+128 updates. Joint denotes JointPairAcc at IoU 0.5. Scores are percentages except same-answer counts.

Backbone	Training	mIoU $\uparrow$	Joint $\uparrow$	Swap error $\downarrow$	Same-answer pairs $\downarrow$
SpotSound-A	Official checkpoint	37.19	0.67	66.33	32/300
SpotSound-A	Paired SFT	76.08	55.33	36.33	61/300
SpotSound-A	Candidate supervision	87.70	81.67	7.00	2/300
TimeAudio	Official checkpoint	40.23	0.00	83.67	67/300
TimeAudio	Paired SFT	38.89	2.67	78.33	160/300
TimeAudio	RelTwin	43.86	10.67	54.33	34/300

swap error falls from 78.33 to 54.33 and identical-answer pairs from 160 to 34.

Both windows contain the queried categories, so improved joint correctness supports occurrence selection beyond category recognition. Figure 1 illustrates SpotSound-A candidate supervision.

Fewer identical answers indicate query sensitivity; higher JointPairAcc confirms correct localization without window swaps. Together, they support query-specific selection beyond answer diversity.

3.4. Binding gains persist after timing intervention

On the 40 recordings with altered silence timing, candidate supervision retains a 20-point JointPairAcc advantage (87.50 vs. 67.50; CI [2.94, 33.33]), compared with original scores of 85.00 and 42.50. The advantage persists when event waveforms, order, and queries are fixed but silence timing changes.

Fixed silence duration or window placement therefore

cannot solely explain the binding gain. Public benchmarks test general localization. The matched ablation isolates complete-answer competition, while temporal recomposition tests whether relation selection survives changed timing. Stage 2 extends this competition from occurrence choice to interval coverage, count, and boundaries.

4. CONCLUSION

RelTwin addresses query-window binding by contrasting acoustically valid timestamp answers with different relational meanings. It improves two backbones on three benchmarks, selects different correct windows for inverse queries, and retains the advantage after silence timing changes. The framework learns relation-aware occurrence selection while preserving direct timestamp generation.

Gains across timestamp representations support transferable answer competition, retaining native inference without candidate scoring or reranking.

5. ACKNOWLEDGEMENTS

Compliance with Ethical Standards. We use public ESC-50 [15], SpotSound [8], Clotho-Moment, UnAV [2], TEMPO, and AudioGrounding data, including programmatically constructed sequences. The study involved no new recordings, participant recruitment, human or animal experiments, or listening studies.

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